



Original Research

Quantifying rill erosion processes in evolving soil landscapes: An experimental approach

Yanmin Jiang^{a, b, c}, Haijing Shi^{a, c, *}, Xihua Yang^{d, e}, Minghang Guo^{a, c}, Weihua Fang^f, Junfeng Shui^{a, c}, Youfu Wu^g, David Paull^h, Ye Wang^{a, c}

^a College of Soil and Water Conservation Science and Engineering (Institute of Soil and Water Conservation), Northwest A&F University, Yangling, 712100, China

^b School of Surveying and Land Information Engineering, Henan Polytechnic University, Jiaozuo, 454000, China

^c Institute of Soil and Water Conservation, Chinese Academy of Sciences and Ministry of Water Resources, Yangling, 712100, China

^d New South Wales Department of Climate Change Energy, the Environment and Water, Parramatta, 2150, Australia

^e School of Life Sciences, University of Technology Sydney, Broadway, 2007, Australia

^f Nanjing Research Institute of Hydrology and Water Conservancy Automation, Ministry of Water Resources, Nanjing, 210000, China

^g Geothermal and Geological Party, Tibet Bureau of Geology and Mineral Exploration and Development, Lhasa, 850000, China

^h School of Science, University of New South Wales Canberra, Canberra, BC, 2610, Australia

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ABSTRACT

Rill erosion is a hydrological-geomorphic process occurring on hillslopes across small spatial extents. Despite advances in soil erosion research, the dynamic feedbacks among the rill erosion process, morphological adaptation, and hydraulic characteristics remain poorly quantified. The current study introduces a high-resolution photogrammetric technology to resolve real-time coupling of rill network evolution with hydraulic characteristics and sediment yield. Four sets of successive rainfall simulation experiments were done with four rainfall intensities of 30, 60, 90, and 120 mm/h. A photogrammetric observation system was applied to dynamically capture the landscape evolutions of rill erosion during ongoing rainfall. The results showed that rainfall of 120 mm/h caused the most damage to soil surfaces with the soil loss being about 9 times higher than that at 30 mm/h. The key morphological indicators of rill density, splitting degree (ω), fractal dimension, and geomorphological contentropy (H) increased with rainfall duration and intensity, while the width-depth ratio decreased. H and ω were the parameters most closely related to sediment yield, contributing 61 % and 39 %, respectively. Based on H and ω , a comprehensive quantitative indicator (CQI) was established by regression analysis. The CQI increased exponentially with the rainfall duration. Among all hydraulic variables, the stream power was the best parameter to characterize the dynamic mechanism of soil erosion and the morphology. These findings are useful for understanding the rill erosion process on a slope and assessing landscape degradation by erosion.

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1. Introduction

Rill erosion is a widely occurring hydrological–geomorphic process on soil-mantled hillslopes and represents one of the most pervasive forms of soil erosion at small spatial scales. It poses

substantial risks to agricultural productivity, ecological restoration, and environmental sustainability (Dai et al., 2020; Hayatzadeh et al., 2023; Li et al., 2024). Rainfall provides the essential source of water for slope runoff, while overland flow serves as the primary driving force behind the initiation and development of rill erosion (Wirtz et al., 2012; Zhang et al., 2009). Through their interaction with the soil surface, rainfall and overland flow detach soil particles and continuously reshape microtopography, making them key variables in eco-hydrological and erosion models. Under heavy or prolonged rainfall, the continuity of overland flow across the microrelief strengthens, accelerating

* Corresponding author. College of Soil and Water Conservation Science and Engineering (Institute of Soil and Water Conservation), Northwest A&F University, Yangling, 712100, China.

E-mail address: shihaijingcn@nwfau.edu.cn (H. Shi).

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the transition from splash or sheet erosion to rill formation (Di Stefano et al., 2013). Once rills emerge, they modify runoff pathways, alter hydrological connectivity, and enhance sediment detachment and transport, thereby driving further rill expansion and morphological evolution (Shen et al., 2016; Zheng et al., 1989). Consequently, improving the understanding of rill erosion processes is essential for eco-hydrological analysis, soil erosion modeling, and erosion control.

With the growing understanding of rill erosion and the advancement of observational technologies, recent studies have increasingly emphasized the quantitative characterization of rill erosion processes (Hussein et al., 2023; Li, Chen, Tao, et al., 2022; Sun et al., 2022). Many researchers have examined rill erosion by monitoring changes in runoff, sediment yield, and the development of soil erosion features (He et al., 2016; Shen et al., 2019; Yan et al., 2024). Runoff and sediment yield remain the most intuitive indicators of slope erosion severity because they are relatively easy to measure and respond directly to hydrodynamic forcing. However, capturing the dynamic evolution of rill morphology is substantially more challenging due to the complexity of the erosion system and the inherent randomness of rill development, particularly under continuous rainfall conditions (Zhang et al., 2019). Current experimental techniques often allow only post-event measurements, resulting in morphological data that represent static end states rather than continuous geomorphic change (Jiang et al., 2020, 2022). As a result, many previous studies simplify the development of rills and lack quantitative descriptions of how rill networks evolve over time.

The hydrodynamic characteristics of overland flow play a fundamental role in determining both the erosion intensity and the morphological development of rills on hillslopes (Lou et al., 2022). Various hydraulic variables, such as flow velocity, the Darcy–Weisbach flow resistance coefficient (Di Stefano et al., 2022; Yang et al., 2022), shear stress, and unit stream power (Atefeh et al., 2022; Gong et al., 2011; Nearing et al., 1997; Nicosia et al., 2023) have been widely used to describe the hydrodynamic processes that drive rill erosion. Many studies have attempted to interpret rill erosion through these hydraulic variables (Niu et al., 2019). For instance, Chen et al. (2011) suggested that variations in rill sediment yield are fundamentally governed by the dynamic behavior of flow shear stress. Nearing et al. (1997) and Zhang et al. (2005) reported that stream power provides a more reliable predictor of soil erosion than shear stress or unit flow stream power, a conclusion also supported by Wang et al. (2006), who found stream power to be the most suitable parameter for modeling soil detachment rates (SDR). Nevertheless, the applicability of these findings varies among studies, as hydrodynamic responses are strongly influenced by experimental conditions such as slope gradient, rainfall intensity, and soil type (Ni et al., 2018). Therefore, clarifying the hydrodynamic mechanisms that govern rill erosion is essential for understanding how rills initiate, evolve, and reshape the soil surface.

Rainfall intensity is a key factor controlling rill erosion because it directly regulates raindrop impact, runoff initiation timing, and runoff erosivity, thereby, influencing the detachment and transport of surface soil particles (Planchon et al., 2000; Yamasaki et al., 2023). As rainfall intensity increases, rill formation becomes more likely (Brunton & Bryan, 2000; Li, Chen, Han, et al., 2022). Li, Chen, Tao, et al., 2022nd soil loss generally increases as well (Berger et al., 2010). However, the relation between rill erosion and rainfall intensity is not linear; instead, rill development is characterized by rapid and complex changes that occur within a single rainfall event (Hussein et al., 2023; Shen et al., 2019; Wirtz et al., 2012). Although previous studies have significantly advanced the understanding of how rainfall intensity affects rill erosion processes (Jiang et al.,

2018; Li, Chen, Tao, et al., 2022; Shen et al., 2016), most experiments relied on rill morphology measured only after rainfall or at a limited number of interruptions during rainfall (Hao et al., 2019; He et al., 2014; Jiang et al., 2018). Such approaches cannot capture the fast-evolving sediment pathways or the temporal progression of rill initiation and bifurcation. These transient adjustments highlight the need for a monitoring technique that provides both high spatial resolution and minute-level temporal resolution—capabilities that traditional static surveys cannot offer.

Dynamic monitoring of rill evolution is essential because key geomorphic adjustments often occur within very short time intervals (on the order of 1–2 min), such as headcut retreat, side-wall collapse, micro-channel bifurcation, and sediment pulse movement. Traditional approaches that obtain only pre- and post-rainfall topography, or low-frequency measurements with long acquisition time, cannot resolve these transient erosion processes nor capture the continuous feedback between hydrodynamics and surface morphology. Although a number of three dimensional (3D) measurement technologies (e.g., laser scanners, light detection and ranging (LiDAR), unmanned aerial vehicle-structure from motion (UAV-SfM)) provide high spatial accuracy, their temporal resolution, operational constraints, and limited applicability in indoor rainfall simulations prevent them from monitoring evolving rill morphology during rainfall. In contrast, digital close-range photogrammetry, which relies on the advantages of fast, flexible, and simple procedures for data acquisition and post-processing, has been extensively applied to erosion monitoring (Leduc et al., 2019). Although digital photogrammetry has been used in generating digital elevation models (DEMs) (Pyle et al., 1997; Rieke-Zapp et al., 2009), how to dynamically monitor the erosion morphology during ongoing rainfall is still an unresolved. To address this issue, the current research group previously developed two digital close-range photogrammetry systems (DCRPS): i) V1.0 avoiding raindrop interference by limiting the field of vision during image acquisition, and ii) V2.0 removing raindrop interference from images to capture real-time evolution morphology in indoor artificial rainfall simulation experiments (Guo et al., 2016; Jiang et al., 2019, 2020, 2022, 2023). Yet, the advantages of photogrammetry to monitor erosion behavior during ongoing rainfall have not been extensively investigated.

To address real-time coupling of rill network evolution with hydraulic characteristics and sediment yield, the current study did four artificial rainfall simulation experiments under different rainfall intensities. A digital close-range photogrammetry system was applied to dynamically monitor rill network development and erosion morphology during ongoing rainfall, enabling continuous capture of rill morphology and overcoming the limitations of conventional static measurements. Specifically, the current study aimed to: (i) discriminate various differences in the processes of runoff and sediment yield under four rainfall intensities; (ii) quantify the morphological characteristics of erosion rills as they evolved with time under different rainfall intensities; (iii) explore the quantitative relation between hydrodynamic characteristics and sediment yield and rill morphology.

2. Material and methods

2.1. Experimental materials

All the experiments were completed in the State Key Laboratory of Soil Erosion and Dryland Farming on the Loess Plateau. A down sprinkler rainfall simulator system covering an area of 27 m × 18 m was used to simulate different rainfall intensities (Zheng & Zhao, 2004). The simulator is installed at a height of 18 m

to ensure raindrops reach terminal velocity and reproduce the kinetic energy and drop-size distribution of natural rainfall (Shen et al., 2015). The system contains 56 grouped nozzles of three diameters (6.35, 12.7, and 19.05 mm), allowing rainfall intensity to be adjusted from 30 to 350 mm/h by regulating water pressure and nozzle configuration (Figs. 1(a), 1(b), and 1(d)). The current experiments were done in a 10 m × 1.5 m × 0.7 m steel soil plot with a collecting device for runoff and sediment at the bottom of the plot (Fig. 1(c)). The soil plot was set at a gradient of 10° to the horizontal, and it had many holes (2 cm aperture) at the bottom to facilitate drainage. The experimental soil was loess, the most typical soil type on the Loess Plateau of China. The loess soil was collected from the hilly region of the Loess Plateau in Ansai County, Shaanxi Province, China. The soil texture was silt loam, composed of 22.1 % sand, 62.0 % silt, and 15.9 % clay.

2.2. Experimental preparation

Before packing the soil plot, the soil was first sieved through a 10 mm sieve to remove stones, plant roots and other debris (Fig. 2(a)), and the soil water content was determined to calculate the soil mass needed to be packed into the soil plot to achieve a target soil bulk density of 1.2 g/cm³ for each layer. The soil moisture was set at about 10 % to be consistent with natural soil (Fig. 2(b)). A 10 cm sand layer was first packed at the bottom of the soil plot to ensure good water permeability. Then, 40 cm of sieved loess was added in 5 cm increments, with each layer was lightly compacted and leveled before adding the next layer (Fig. 2(c)). Before the formal experiment, the soil plots were subjected to a low-intensity rainfall (30 mm/h) to pre-wet the soil surface until surface flow occurred (Momm et al., 2018). After pre-wetting, the soil surface was covered with a plastic sheet and rested for 24 h to allow moisture redistribution.

To compare erosion responses across rainfall intensity gradients representative of common erosive events on the Loess Plateau (Zhou & Wang, 1987, 1992), four intensities (30, 60, 90, and 120 mm/h) were applied in the current study. Each treatment was



Fig. 2. Preparation and procedures of the simulated rainfall experiments.

repeated twice. Before each run, to calibrate the accuracy of rainfall intensity, eight buckets were uniformly placed around the soil plot to collect the rain until the difference between the measured rainfall intensity and the desired rain intensity was less than 5 % (Zhang et al., 2016). Moreover, the rainfall uniformity was assessed using the Christiansen coefficient uniformity approach (Mahmoodabadi et al., 2007). The results showed that the rainfall intensities were 30.89, 59.5, 90.8 and 119.22 mm/h, and the rainfall uniformity coefficients were 96 %, 96 %, 94 %, and 92 %, respectively.

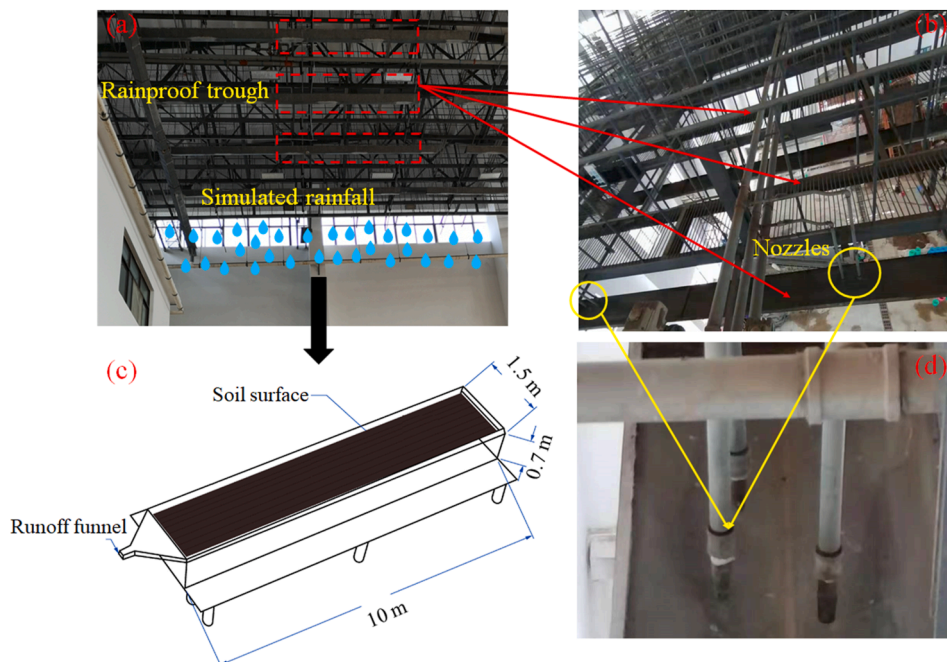


Fig. 1. Rainfall simulator and soil plot setup. (a) Bottom view of the rainfall simulator, (b) top view of the rainfall simulator, (c) schematic of the experimental soil plot, and (d) nozzles.

2.3. Experimental procedure

Prior to each run, the undisturbed initial soil surface morphology was first measured using photographic observation technology for comparison with the eroded surface. During rainfall application, runoff and sediment samples were collected from the plot outlet every minute after runoff occurred. Meanwhile, digital images of the eroded surface landscape were acquired using the DCRPS at 5-min intervals. The soil surfaces at four rainfall intensities were all subjected to simulated rainfall for about 60 min after the appearance of slope runoff. During each image acquisition, six targets were placed around the soil plot as the dimensional constraint in the vertical and horizontal directions. The 10 m long slope was divided into 0–2, 2–4, 4–6, 6–8, and 8–10 m sections. Surface runoff velocity was measured every 5 min at each segment using the potassium permanganate (KMnO₄) dye-tracing method (Fig. 2(d)). The runoff velocity for each section was multiplied by a correction factor of 0.75 as the correction factor value (Dong et al., 2014; Luk & Merz, 1992), and the average runoff velocity used in this study was the average at the five sections of the slope (Table 1). The water temperature for the four rainfall intensities was approximately 10 °C, measured using a thermometer.

2.4. Data processing and analysis

2.4.1. Runoff and soil loss

Once runoff commenced, sediment samples were collected at 1-min intervals to ensure the accuracy of the measured runoff volume and sediment mass. A 1000-mL bottle was first used to collect runoff and sediment samples, and the remaining outflow was collected in a runoff bucket. After each rainfall event, the runoff and sediment samples were weighed and left to settle for 24 h. The supernatant was then decanted, and the remaining sediment was transferred to aluminum boxes, oven-dried at 105 °C, and weighed to determine the sediment yield (S).

2.4.2. DEM generation and rill network extraction

The DCRPS was instrumented with 12 Sony complementary metal oxide semiconductor cameras (AMOS) (20 mm fixed lens) (Fig. 3). The cameras were positioned on a steel frame 14 m above the ground to cover the entire bed surface (Jiang et al., 2020, 2023), and were networked through a wireless router to realize the concurrent operation of image acquisition. The coverage of each camera was a block, and there were overlapping images between adjacent blocks. Each camera continuously collected 60 images at a time, and the image overlap between adjacent blocks was

30 %–45 %. Thus, before constructing DEMs, the images of soil surface morphology were first transformed into digital point clouds using a series of data processing procedures in DCRPS. The DCRPS achieved a horizontal accuracy of approximately 2 mm and a vertical accuracy of approximately 5 mm, as validated in Jiang et al. (2023). This level of precision is sufficient for resolving fine-scale variations in rill morphology. In addition, although small uncertainties exist in DEM generation, the overall precision of the photogrammetric system is sufficient for the level of morphological detail required in the current study, and potential errors do not affect the conclusions regarding rill development trends or the quantitative relation between morphology, hydrodynamics, and sediment yield.

The processing procedures included raindrop removal, image transmission, point cloud matching, point cloud patching, and 3D reconstruction. All steps were remotely controlled by a host computer (Fig. 3) and are described in detail in Section 2 of Jiang et al. (2023). After processing, DEMs were generated using the 3D Analyst tools in ESRI ArcGIS 10.2. Prior to DEM construction, outliers and noise points in the digital point clouds were removed through visual inspection in Cyclone 6.0.3 (Wu et al., 2018). The resulting DEMs had a spatial resolution of 2 mm × 2 mm. Rill networks were extracted from the DEMs using the hydrological analysis module in ArcGIS based on the D8 (eight-direction) flow algorithm (O'Callaghan & Mark, 1984). Specifically, depressions in the DEM were filled using the Fill tool, followed by calculation of flow direction. The Flow Accumulation tool was then used to compute the amount of contributing flow (i.e., the number of upstream cells draining to each cell). Based on the flow accumulation results, an appropriate threshold was selected to identify and extract the rill network. Once the rill network was delineated, the rill morphological characteristics of rill density, splitting degree, width-depth ratio, fractal dimension, and geomorphological comentropy were quantified using spatial Analyst tools in ArcGIS and Python scripts. The specific technical flow chart is shown in Fig. 4.

2.4.3. Rill morphological parameters

The rill density (ϵ) is a measure of the extent to which the hillslope is fragmented by water flow, calculated as the total rill length divided by the total study area (m/m²) (Bewket & Sterk, 2003). A higher rill density indicates a greater degree of fragmentation of the slope. The calculation formula is as Eq. (1):

$$\epsilon = \sum_{i=1}^n L_{ti} / A_0 \quad (1)$$

Table 1
Average flow velocity and depth at different times.

Time (min)	Flow velocity (m/s)				Flow depth (mm)			
	30 mm/h	60 mm/h	90 mm/h	120 mm/h	30 mm/h	60 mm/h	90 mm/h	120 mm/h
5	0.12	0.15	0.19	0.23	1.11	2.18	3.49	4.51
10	0.13	0.16	0.21	0.25	1.37	2.58	3.67	4.68
15	0.13	0.15	0.23	0.27	1.91	2.74	3.67	4.93
20	0.15	0.16	0.23	0.27	1.83	3.05	3.91	4.95
25	0.15	0.16	0.24	0.25	1.88	3.15	3.91	5.34
30	0.15	0.17	0.22	0.26	1.96	2.73	4.02	5.40
35	0.15	0.17	0.22	0.27	2.04	3.14	3.95	5.21
40	0.16	0.17	0.23	0.28	2.03	3.06	4.01	5.22
45	0.16	0.18	0.24	0.27	2.05	3.09	3.45	6.08
50	0.15	0.18	0.24	0.28	2.24	3.40	3.69	6.10
55	0.14	0.19	0.25	0.29	2.57	3.25	4.09	5.69
60	0.15	0.19	0.26	0.28	3.84	4.79	5.05	6.00

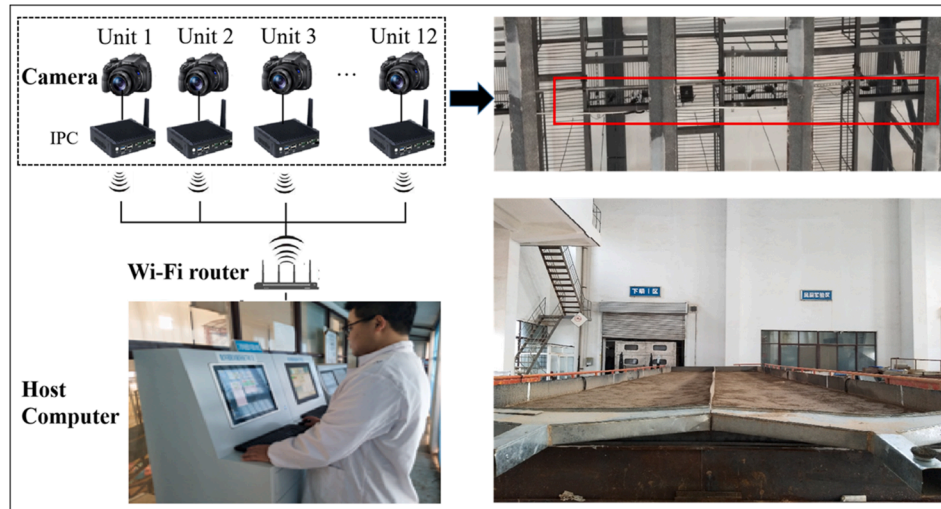


Fig. 3. Wireless networking for the photogrammetric observation system.

where A_0 is the surface area (m^2) of the studied slope; L_{ti} is the total length (m) of the i th rill and its bifurcation on the slope; i is the rill number, $i = 1, 2, \dots$; o is the total number of rills.

The rill splitting degree (ω) is the total planar area of all rills divided by the total study area, which reflects the degree of slope fragmentation and the intensity of rill erosion. The expression is as Eq. (2):

$$\omega = \sum_{i=1}^o A_i / A_0 \quad (2)$$

where A_i is the surface area of the i th rill on the slope (m^2).

Rill width-depth ratio ($R_{w/d}$) is the ratio of the rill width to its corresponding depth, which describes the shape of the rill cross

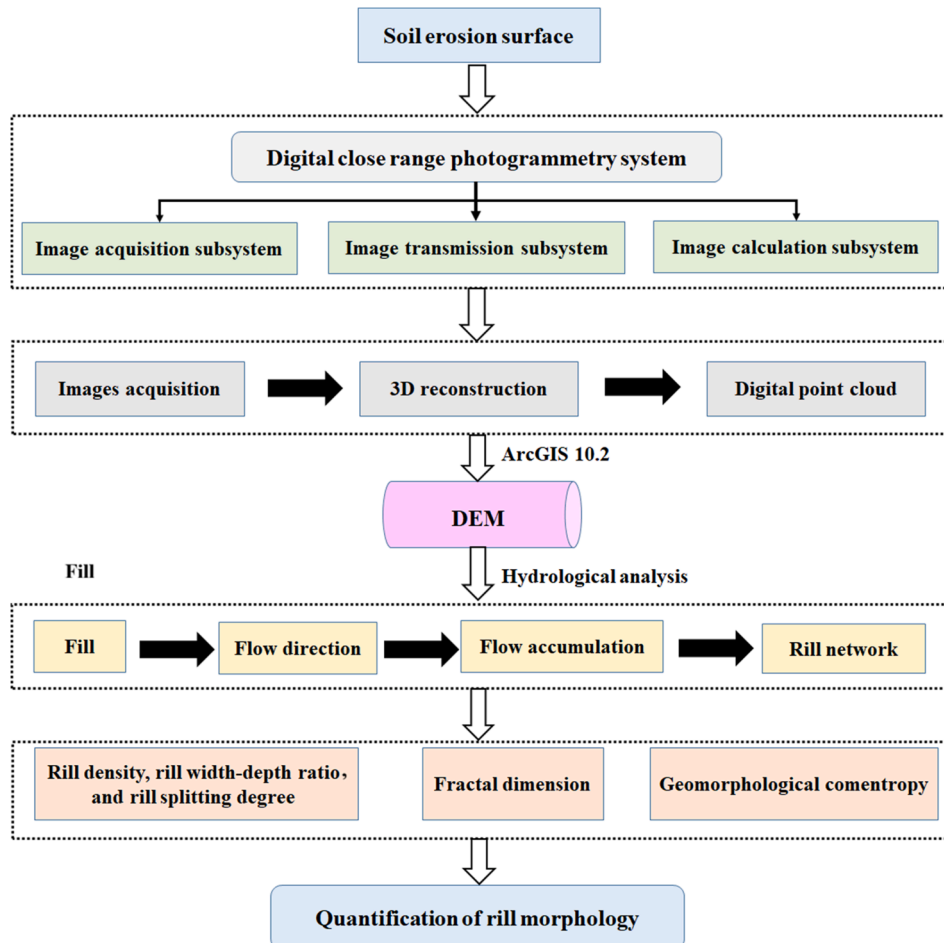


Fig. 4. Technical flow chart of DEM generation and rill network extraction.

section. This parameter is dimensionless. A large $R_{w/d}$ value indicates that the cross-section shape of the rill is wide and shallow. Inversely, the lower $R_{w/d}$ value denotes a narrow and deep rill cross-section shape. The calculation formula is expressed as Eq. (3):

$$R_{w/d} = \frac{\sum_{j=1}^z w_j}{\sum_{j=1}^z d_j} \quad (3)$$

where w_j is the width of the rill at the j th monitoring point (m); d_j is the depth of the rill at j th monitoring point (m); z is the total number of monitoring points on the slope.

The fractal dimension (D_0) describes the organization of the rill network and its complexity degree (Zhang et al., 2017). Increasing values of D_0 are indicative of a rill network that tends to be complex and mature. Conversely, decreasing values of D_0 denote a simple and young rill network. The fractal dimension is calculated as Eqs. (4) and (5):

$$N(r) = Gr^{-D_0} \quad (4)$$

$$D_0 = \ln N(r) / \ln(1/r) \quad (5)$$

where $N(r)$ is the number of boxes covered by rills; r (mm) is the box size; G is a constant.

Geomorphological comentropy (H) is an entropy-based metric used to quantify the complexity, heterogeneity, and spatial organization of evolving land-surface morphology. Following Ai (1987), H describes the degree of disorder in the distribution of rill-related terrain units and reflects the progressive organization of rill networks during erosion, which is increasingly utilized in soil erosion studies. Entropy characterizes the level of rill development, which gradually increases with the evolution of rills. Although “comentropy” is less commonly used than landscape entropy or fractal complexity, it is conceptually aligned with these established indicators because all measure the degree of structural disorder within a geomorphic system. The specific calculation procedures are as follows: first, the DEM of each erosion landform was classified by elevation, and drainage area above different elevations was calculated. The elevation curve function of $y = f(x)$ with the best fitting effect was then obtained by fitting the area ratio (A') and the elevation ratio (H') in the basin (Jiang et al., 2024). Finally, the Strahler area-elevation integral value was obtained by integrating the function. A Python code has been developed for DEM reading and the geomorphological comentropy calculation on the VC+6.0 platform. The geomorphological comentropy was expressed as Eqs. (6) and (7):

$$S = \int_0^1 f(x) dx \quad (6)$$

$$H = S - 1 - \ln S = \int_0^1 f(x) dx - 1 - \ln \left[\int_0^1 f(x) dx \right] \quad (7)$$

where S is the Strahler area-elevation integral value; $f(x)$ is Strahler's hypsometric curve (Strahler, 1952).

The A' and H' are dimensionless parameters defined as Eqs. (8) and (9),

$$H' = \frac{E - E_{\min}}{E_{\max} - E_{\min}} \quad (8)$$

$$A' = \frac{A}{A_{\text{total}}} \quad (9)$$

where E is the elevation at a given point; $E_{\max}$ and $E_{\min}$ are the maximum and minimum elevations within the slope, respectively; A is the cumulative area at a given elevation; A_{total} is the total area of the erosion slope.

2.4.4. Hydrodynamic parameters

Hydrodynamic parameters are indispensable factors for the analysis of rill morphology evolution and runoff and sediment yield of slopes, and they are crucial for comprehending the hydrodynamic mechanism and erosion process (Cao et al., 2015). In the current study, the commonly used hydrodynamic parameters of flow velocity (V), Froude number (Fr), Reynolds number (Re), Darcy–Weisbach resistance coefficient (f), unit stream power (U), stream power (W), and flow shear stress (τ) were selected to analyze the effects of rainfall intensity on hydrodynamic characteristics. The calculation expressions of Fr , Re , and f were described in detail in Section 2.3.3 of Jiang et al. (2022).

Moore and Burch (1986) presented the concept of unit stream power according to the equation for sediment transport. Unit stream power is calculated as Eq. (10):

$$U = VJ \quad (10)$$

where U is unit stream power (m/s); V is the average flow velocity (m/s); J is the hydraulic gradient (m/m).

Flow shear stress is the stress exerted by runoff that causes soil particle separation and sediment transport, and it is expressed as Eq. (11) (Foster et al., 1984):

$$\tau = pgRJ \quad (11)$$

where τ is shear stress (Pa); p is the mass density of the water–sediment mixture (kg/m^3); g is the acceleration due to gravity (9.8 m/s^2); R is the hydraulic radius (m).

Bagnold (1966) reported a close relation between stream power and shear stress, which referred to the power consumed by the flow acting on a unit area. The specific formula is as Eq. (12):

$$W = V\tau \quad (12)$$

where W is stream power ($\text{N/m}^2/\text{s}$).

2.4.5. Grey relational analysis

Grey relational analysis (GRA) is a quantitative method used to evaluate the degree of association between variables based on the similarity of their development trends (Lyle & Smerdon, 1965). As a mature technique for analyzing systems with limited samples and uncertain information, GRA has been widely applied in hydrology and soil erosion studies. Given the stochastic nature of rill morphological evolution and the difficulty in determining the direct response between morphological indices and sediment yield, GRA is particularly suitable for the current study. In this study, the measured sediment yield series was selected as the reference sequence, while rill morphological parameters were treated as comparison sequences. Following standard GRA procedures (normalization, calculation of grey relational coefficients, and computation of relational grades), the correlation strength between each morphological parameter and sediment yield was obtained. The computational flow chart is shown in Fig. 5.

3. Results

3.1. Variation characteristics in runoff and sediment yield at different rainfall intensities

The variations in runoff rate and cumulative runoff during the 60-min runoff period were similar across the different rainfall intensities, whereas the total runoff volume after rainfall differed significantly (Figs. 6(a) and 6(c)). The overall trends show that both runoff rate and cumulative runoff increased with rainfall duration and rainfall intensity. The initiation times of runoff under rainfall intensities of 30, 60, 90, and 120 mm/h were 11.35, 8.27, 3.61, and 1.89 min, respectively (Table 2). Increasing rainfall intensity resulted in earlier runoff initiation and greater runoff volume; for example, runoff under rainfall at 120 mm/h occurred approximately 10 min earlier than that under 30 mm/h, and the total runoff volume at 120 mm/h was nearly four times that at 30 mm/h.

The sediment yield rates under the four rainfall intensities exhibited two distinct patterns (Fig. 6(b)). Under rainfall intensities of 30 and 60 mm/h, the sediment yield rate initially increased and then decreased during the early stage of the experiments, followed by a sudden surge after fluctuating at relatively low values. In contrast, under 90 and 120 mm/h, the sediment yield rate first decreased from a high initial value and then increased with fluctuations throughout the rainfall process. Overall, the sediment yield rate increased with runoff rate following a power-law relation (Fig. 6(e)), indicating that larger runoff rates transported more sediment. Consistent with this, higher rainfall intensity produced greater cumulative sediment yield (Fig. 6(d)), with the maximum sediment yield and sediment transport rate reaching 104.4 kg and 116.0 g/m²/min, respectively. Additionally, cumulative sediment yield after 40 min of runoff increased exponentially with cumulative runoff (Fig. 6(f)).

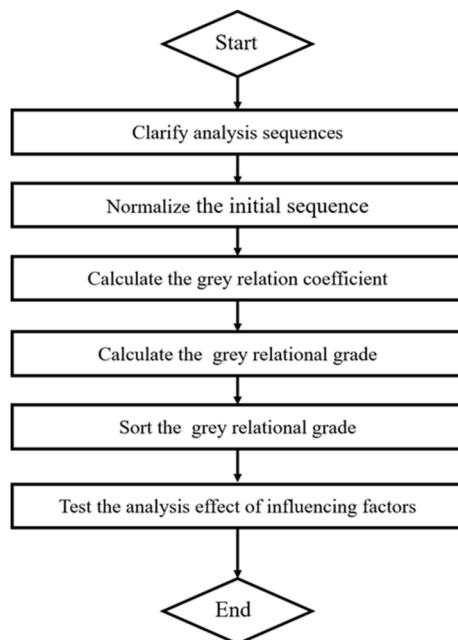


Fig. 5. Flow chart of the grey relational analysis (GRA).

3.2. Rill erosion morphology

3.2.1. Erosion landscape evolution

Fig. 7 shows the digital point clouds acquired at different time intervals. Based on the erosion traces observed on the evolving soil surface, rill development followed a typical geomorphic sequence: splash erosion, sheet erosion, erosion pit formation, rill initiation, and rill network development. The duration of each erosion stage varied under different rainfall intensities, and the degree of slope degradation increased progressively with rainfall intensity.

At rainfall intensities of 30 and 60 mm/h, rills developed slowly, and changes in erosion patterns were relatively subtle. Under 30 mm/h, initial rills did not appear until 31 min after rainfall onset (Table 2), and these rills were located near the bottom of the slope. At 60 mm/h, the first rills appeared at 17.43 min, emerging in the upper-middle portion of the slope. In contrast, rill morphology changed markedly throughout the rainfall events at higher rainfall intensities of 90 and 120 mm/h. The developmental processes at these two intensities were similar. Initially, two small and discontinuous erosion pits formed on the slope surface, from which rills subsequently developed. These rills extended downslope as rainfall continued. With further erosion, the lower rills gradually retreated upslope, moving toward the upper rills. By the end of the rainfall event, the two rills under 90 mm/h remained unconnected, whereas under 120 mm/h they merged, forming a continuous rill network.

At 30 and 60 mm/h, the rill networks remained young and insufficiently developed during the experiments. Under 90 and 120 mm/h, two discontinuous rills emerged and extended along the slope direction. As erosion progressed, these rills expanded laterally along both sides of the slope and gradually approached each other. Eventually, the upper and lower rills connected to form a continuous channel, and a complex rill network with a dense distribution of slope rills was established.

3.2.2. Quantitative indicators of rill morphology

Table 3 lists the quantitative indicators of rill morphology at a 10-min interval after the rills appear at 30, 60, 90, and 120 mm/h rainfall intensities. Except for the rill width-depth ratio, rill density, splitting degree, H , and fractal dimension all increased with rainfall duration and intensity, but the increases for them varied between the four rainfall intensities. Regarding rainfall at 30 and 60 mm/h, rills began to develop after rill embryos appeared, however, the increase in morphological parameters was not significant due to the slow development rate. The splitting degree was 0.012 and 0.024 at the end of rainfall, respectively. Compared with 30 and 60 mm/h, rills developed rapidly after 20 min at 90 and 120 mm/h, and the morphological parameters significantly changed, especially rill density. The rill density increased by 2.459 and 2.977 m/m² from 20 to 60 min, respectively. In contrast, the rill networks formed after rainfall at 30 and 60 mm/h were still young with the rill densities of 0.419 and 0.619 m/m², respectively. Furthermore, from the variation process of width to depth ratio, the increase in rill width was smaller than that of rill depth, especially for rain of 30 mm/h. The change in fractal dimension denotes that the rill network had obvious fractal characteristics.

3.2.3. Geomorphological comentropy

Entropy reflects the degree of rill development, with rill evolution corresponding to an increase in entropy. The entropy value, H , of geomorphic information at different time intervals was calculated using Eq. (4), and the results are listed in Table 3. For each rainfall intensity, H increased with rainfall duration, and it also increased with increasing rainfall intensity (Fig. 8(b)). The

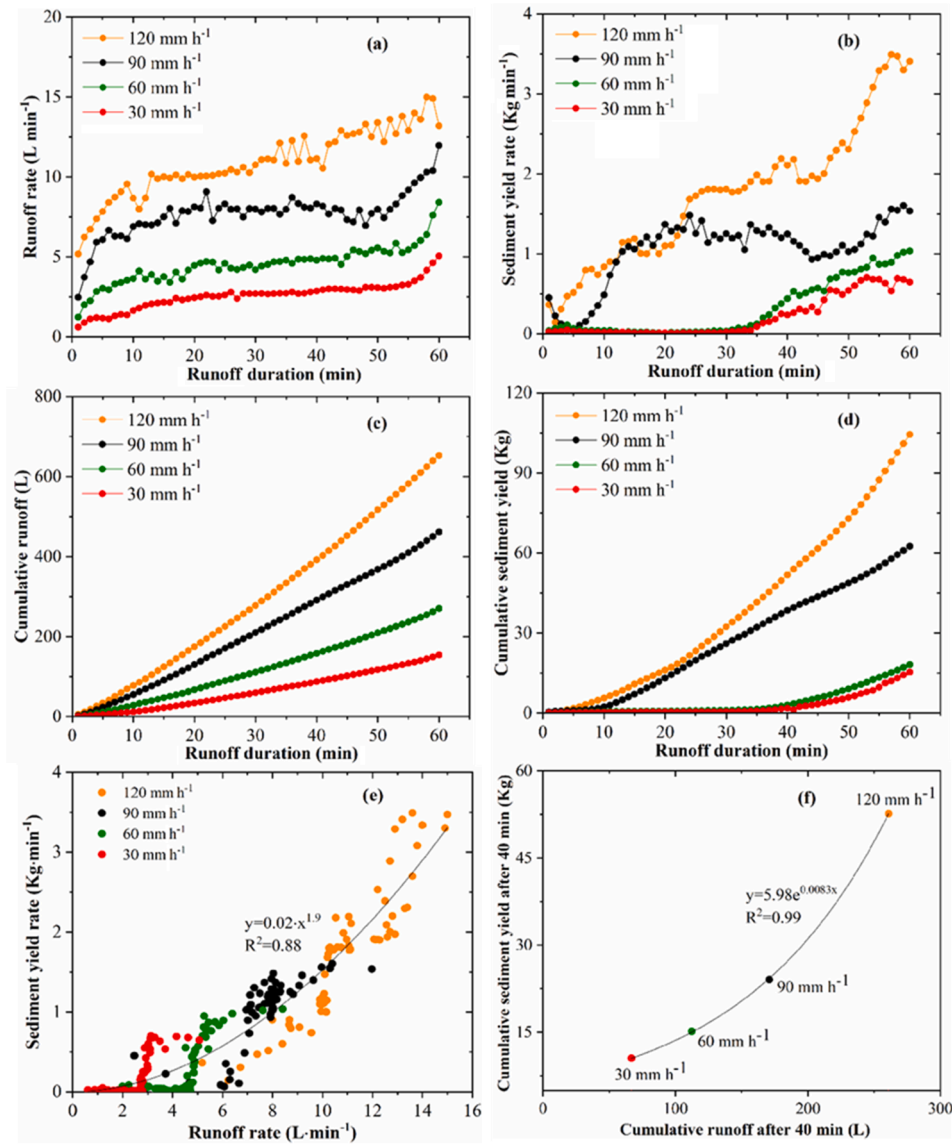


Fig. 6. Temporal variation of sediment yield and runoff volume at different rainfall intensities.

final H values under rainfall intensities of 30, 60, 90, and 120 mm/h were 0.207, 0.224, 0.330, and 0.379, respectively. The largest H value occurred under 120 mm/h, indicating the most active erosion processes and the most severely disturbed terrain.

The results show that the larger the integral area S , the smaller the corresponding entropy H . Under the rainfall intensity of 30 mm/h, the soil erosion landscape exhibited the largest integral area and the smallest H value. Fig. 8(a) shows the area–elevation integral curves of the erosion landforms following rainfall at the

four rainfall intensities (30, 60, 90, and 120 mm/h), and the fitted area–elevation curve functions are as Eqs. (13)–(16).

$$\begin{cases} y_{30} = 1.0439 - 1.9501x + 2.2788x^2 - 1.3697x^3, R^2 = 0.995 \\ y_{60} = 0.0970 - 2.0105x + 2.899x^2 - 1.8012x^3, R^2 = 0.998 \\ y_{90} = 1.0129 - 1.9384x + 2.7136x^2 - 1.6077x^3, R^2 = 0.997 \\ y_{120} = 0.9302 - 2.5935x + 3.9043x^2 - 2.1005x^3, R^2 = 0.995 \end{cases} \quad (13-16)$$

Table 2

Initiation time of runoff and rill formation at different rainfall intensities.

Rainfall intensity (mm/h)	30	60	90	120
Time to runoff initiation (min)	11.35	8.27	3.61	1.89
Time to rill initiation (min)	31.00	17.43	13.19	12.45
Total soil loss (kg)	15.38	18.17	62.5	104.4
Total runoff volume (L)	158.17	270.38	461.79	652.69
Average runoff rate (L/min)	3.72	4.51	7.7	12.77
Average sediment transport rate (g/(m ² min))	16.27	20.19	69.44	116.00

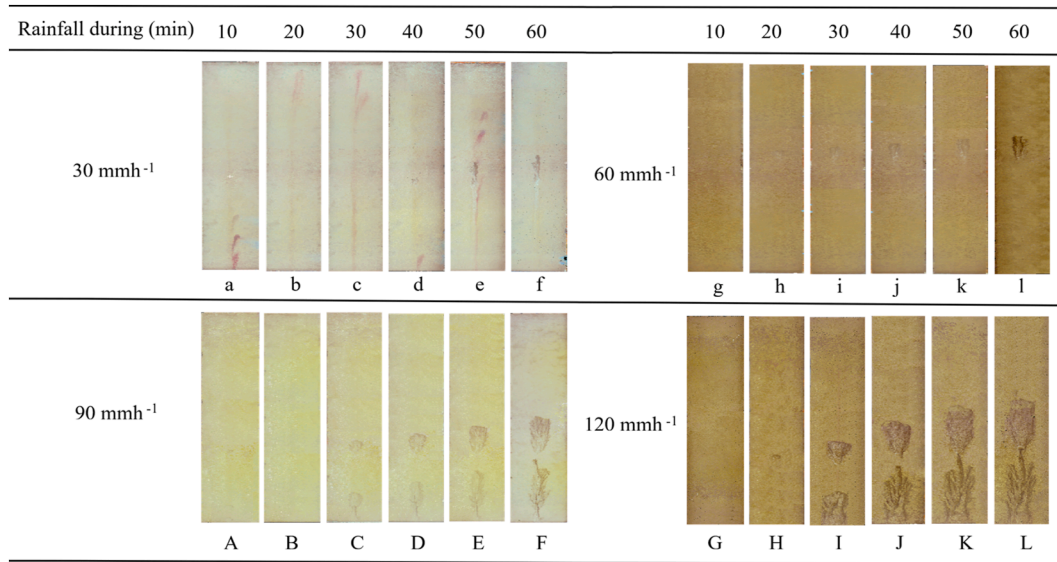


Fig. 7. Digital point clouds at different time intervals in the four rainfall experiments.

Table 3

Rill morphological indicators at different rainfall intensities.

Rainfall intensity (mmh)	Rainfall duration (min)	ε (m/m ²)	ω	$R_{w/d}$	D_θ	H
30	10	—	—	—	—	—
	20	—	—	—	—	—
	30	—	—	—	—	—
	40	0.266	0.003	3.031	1.215	0.189
	50	0.371	0.009	2.280	1.224	0.202
	60	0.419	0.012	2.060	1.228	0.207
60	10	—	—	—	—	—
	20	0.329	0.005	3.732	1.310	0.193
	30	0.393	0.009	3.581	1.394	0.210
	40	0.466	0.015	3.308	1.412	0.215
	50	0.541	0.020	3.150	1.432	0.219
	60	0.619	0.024	2.861	1.443	0.224
90	10	—	—	—	—	—
	20	0.735	0.051	3.482	1.402	0.215
	30	1.351	0.066	3.271	1.421	0.233
	40	1.995	0.080	2.908	1.454	0.279
	50	2.542	0.089	2.671	1.483	0.301
	60	3.194	0.103	2.390	1.489	0.330
120	10	—	—	—	—	—
	20	0.890	0.054	3.336	1.324	0.233
	30	1.714	0.082	3.091	1.470	0.271
	40	2.419	0.109	2.765	1.498	0.328
	50	3.104	0.147	2.422	1.522	0.356
	60	3.867	0.167	2.160	1.533	0.379

3.2.4. Relation between rill morphology and fractal dimension

The fractal dimension of the rill networks under different rainfall intensities was calculated at 10-min intervals using the ArcGIS-based grid method proposed by Zhang et al. (2014). The fitted relation between box size and the number of non-empty boxes at different rainfall durations showed that the logarithms of these two variables exhibited a strong linear relation, with correlation coefficients greater than 0.99 (Fig. 9). The stable ratio between the logarithmic values, regardless of box size, indicates clear fractal characteristics of the rill networks on the slope. Therefore, the slope of the fitted linear function was taken as the fractal dimension of each rill network.

The fractal dimensions increased slightly with rainfall duration for all four rainfall intensities. Across the experimental period, the

fractal dimension values ranged from 1.215 to 1.228 under 30 mm/h, 1.310 to 1.443 under 60 mm/h, 1.402 to 1.489 under 90 mm/h, and 1.324 to 1.533 under 120 mm/h (Table 3).

3.2.5. Relation between morphological indicators and sediment yield

Although the morphological indicators changed with increasing sediment yield, the magnitude of these changes differed among indicators, making it difficult to assess the compactness of the relation between sediment yield and individual morphological indicators. The strength of correlation between rill morphological indicators and sediment yield decreased in the order: $\omega > H > D_\theta > R_{w/d}$ (Table 4). Both ω and H showed strong correlations with sediment yield, indicating that they are the most

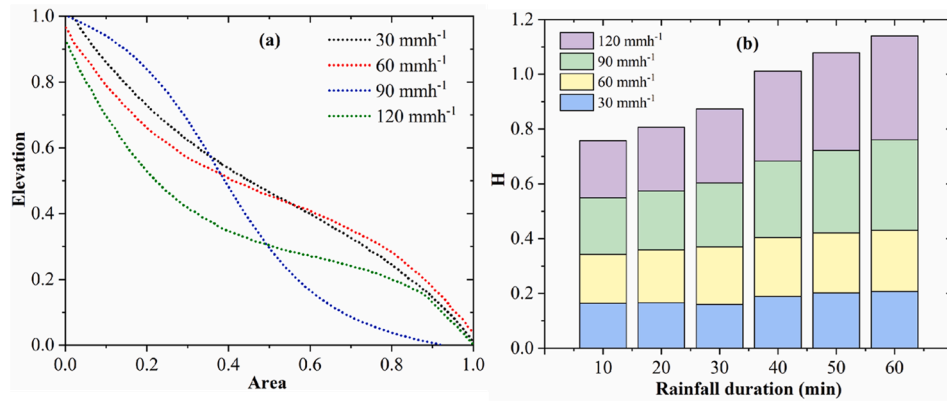


Fig. 8. (a) Dimensionless area-elevation curves and (b) geomorphic information entropy (H) over time for different rainfall intensities (30, 60, 90, and 120 mm/h). Note: Area and elevation were made dimensionless by dividing by the maximum elevation and total area, respectively.

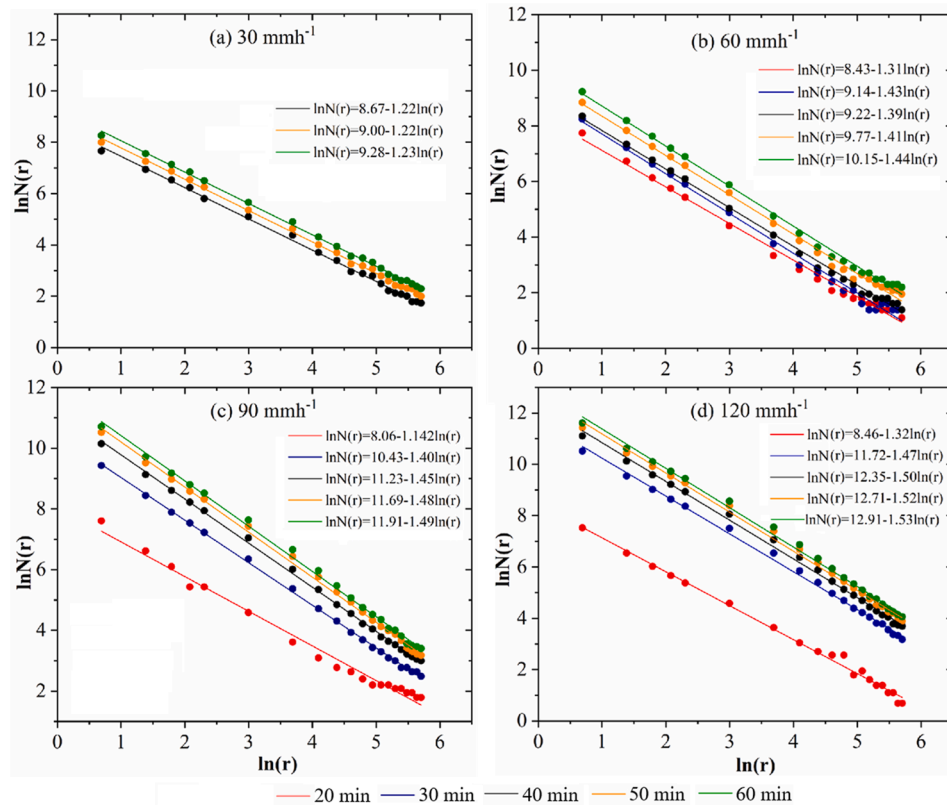


Fig. 9. Relation of fractal dimension and rill networks. Different colored lines represent different durations of rainfall.

Table 4
Grey relational coefficient between morphological indicators and sediment yield.

Morphological parameter	Grey relational coefficient	Rank
ε	0.717	4
ω	0.906	1
$R_{w/d}$	0.692	5
D_δ	0.728	3
H	0.769	2

sensitive indicators for characterizing the degree of erosion development. In contrast, the correlations of D_δ and $R_{w/d}$ with sediment yield were relatively weak.

To determine the comprehensive influence of rill morphology evolution on slope sediment yield, the sediment yield was taken as the objective function, and the morphological indicators ω and H , which were most closely related to sediment yield, were selected as independent variables to establish a multiple linear regression relation. The regression equation is as Eq. (17):

$$S_y = 0.6H + 0.388\omega \quad (R^2 = 0.957, p < 0.05, N = 32) \quad (17)$$

where S_y is the sediment yield (kg), R^2 is the coefficient of determination, p is the probability of obtaining test results at least as extreme as the current results, and N is the number values compared.

The results of regression analysis showed that the P values of constants H and ω were 0.00, 0.00, and 0.011, respectively, indicating that H and ω were significant to the sediment yield. The standardized regression coefficients in Eq. (17) were positive, which indicates that H and ω had positive effects on the average sediment yield. The contribution rate of ω to S_y was 39 %, and of H to S_y was 61 %, the influence of geomorphological comentropy of the slope on the sediment yield was greater than that of the splitting degree, these contribution rates also reflecting the different sensitivities of ω and H to hydrodynamic forcing. The change in rill morphology directly affects the sediment yield of the slope; the more complex the rill network, the greater the sediment mass produced on the slope (Zhang et al., 2017).

Due to the synchronous variation between sediment yield and rill morphology, nearly all eroded sediment was transported to the outlet during the erosion process. Since both H and rill ω show strong correlations with sediment yield, their relation with sediment production can also be used to characterize rill morphology. Therefore, following Ni et al. (2018), a dimensionless parameter known as the comprehensive quantitative indicator (CQI) was used to describe rill morphology. This index is closely related to sediment yield and integrates the effects of H and ω . Based on regression analysis, the CQI is defined by the following multiple linear regression equation (Eq. (18)):

$$\text{CQI} = 0.61H + 0.39\omega \quad (18)$$

The change characteristics of rill morphology with rainfall intensity were quantified using Eq. (18) as the CQI of rill morphology. The CQI increased exponentially with the rainfall duration, which indicated that the complexity of rill morphology gradually increased, but the slope of the fitted curve gradually decreased with time (Fig. 10). The CQI values varied from 0.114 to 0.292, and the correlation coefficients increased with rainfall intensity. Furthermore, regarding the rains at 60, 90, and 120 mm/h, the power exponent gradually increases with rainfall intensity, while the curvature of the inward concave curve decreased.

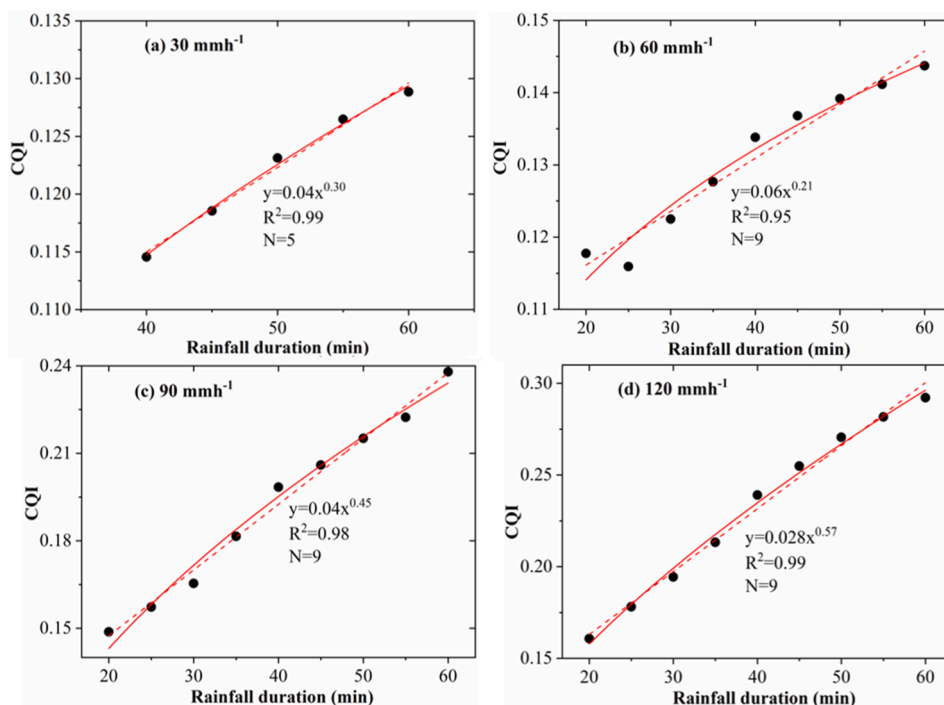


Fig. 10. Variation of CQI with rainfall duration under different rainfall intensities of (a) 30 mm/h, (b) 60 mm/h, (c) 90 mm/h, and (d) 120 mm/h.

3.3. Hydraulic and hydrodynamic characteristics of rill erosion

3.3.1. Flow hydraulic and hydrodynamic characteristics

Fig. 11 shows the temporal variation of hydraulic and hydrodynamic parameters (HHP) under different rainfall intensities. The Froude number (Fr) and Darcy–Weisbach resistance coefficient (f) ranged from 1.27 to 2.25 and 0.30 to 0.86, respectively, and exhibited two distinct patterns across the four rainfall intensities. The overall trend of Fr decreased under 30 and 60 mm/h, whereas it increased under 90 and 120 mm/h (Fig. 11(a)). The parameter f showed an opposite trend to Fr (Fig. 11(b)). In all cases, Fr values exceeded 1, indicating that the flow was supercritical.

The flow unit stream power (U) and Reynolds number (Re) showed similar temporal changes, ranging from 0.015 to 0.037 m/s and 14.3 to 173.9, respectively, and both increased progressively with rainfall intensity and duration (Figs. 11(c) and 11(d)). Stream power (W) varied between 0.05 and 2.61 N/m/s and also displayed two distinct variation patterns across the rainfall intensities. At 30 and 60 mm/h, W increased initially and then decreased. In contrast, at 90 and 120 mm/h, W first decreased and then began to increase sharply after approximately 25 and 10 min, respectively (Figs. 11(e) and 11(f)). Additionally, shear stress (τ) ranged from 0.41 to 13.11 Pa and exhibited a temporal trend similar to that of W under all four rainfall intensities.

3.3.2. Correlation of flow HHP and rill erosion and rill morphology

To investigate the driving mechanism of rill development, the Pearson correlation coefficient was used to examine the relation between HHP and rill morphological parameters. For the association between HHP and the soil detachment rate (SDR), GRA was applied because it is more suitable for nonlinear processes and avoids the multicollinearity inherent among hydrodynamic parameters. The results of correlation analysis showed that HHP was significantly correlated with almost all morphological indicators of the slope at the 0.01 significance level (2-tailed) (Fig. 12). An exception was that it was not significantly correlated with Fr and f .

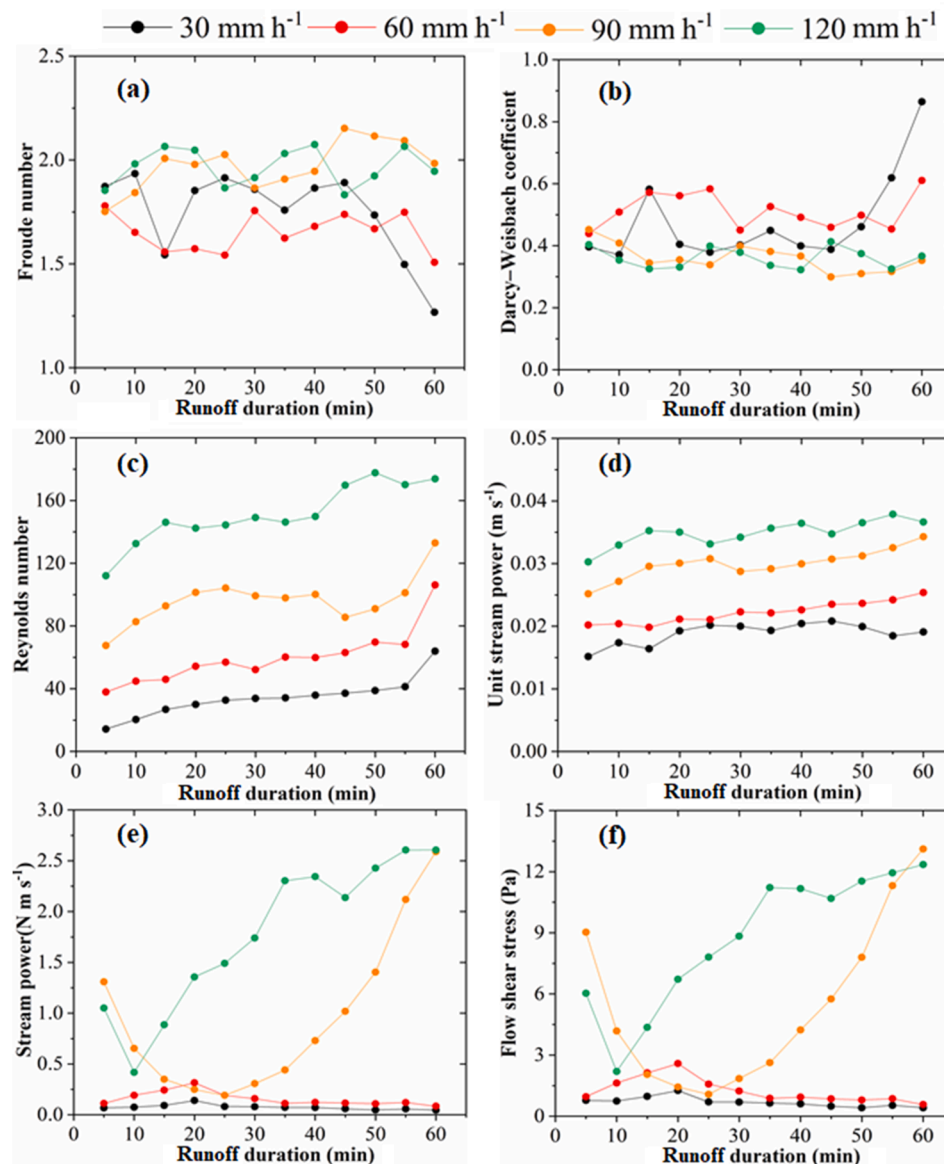


Fig. 11. Variation in flow hydraulic and hydrodynamic parameters with rainfall duration.

and $R_{w/d}$ was negatively significantly correlated with all other parameters. Moreover, f was significantly negatively correlated with the other parameters except for $R_{w/d}$. Among all the HHP, W was the most closely related parameter to the morphological indicators. According to the results of the GRA between HHP and SDR in Table 5, the correlation coefficients decreased in the order of $W > \tau > Re > V > U > Fr > f$. Thus, W is the best hydrodynamic parameter to describe the development trend of SDR, and SDR increased with W exponentially with $R^2 = 0.95$ (Fig. 13).

4. Discussion

4.1. Effects of rainfall intensity on runoff and sediment yield

The current results demonstrate that both the runoff rate and sediment yield increase with rainfall intensity, consistent with earlier findings (Berger et al., 2010; Boardman & Poesen, 2006). Higher rainfall intensity provides greater kinetic energy for soil detachment and enhances the transport capacity of overland flow.

This is supported by the current measurements, which show an exponential increase in sediment yield rate with increasing runoff rate (Fig. 6(e)), in agreement with Berger et al. (2010). Similarly, cumulative sediment yield increased exponentially with cumulative runoff (Fig. 6(f)), confirming the nonlinear nature of runoff–sediment interactions.

Differences in the timing of runoff initiation further illustrate the role of rainfall intensity in controlling erosion onset. Under lower rainfall intensities (30 and 60 mm/h), runoff initiated later due to lower infiltration and the longer time required for the soil to reach saturation. This observation aligns with previous studies indicating that lower rainfall intensities promote a greater proportion of rainfall taken up by initial infiltration, thereby, delaying both runoff and erosion (Planchon et al., 2000; Yamasaki et al., 2023). In contrast, higher rainfall intensities (90 and 120 mm/h) led to much earlier runoff initiation, as the soil's infiltration capacity was rapidly exceeded, resulting in immediate surface runoff and the early onset of erosion (Romkens et al., 2002).

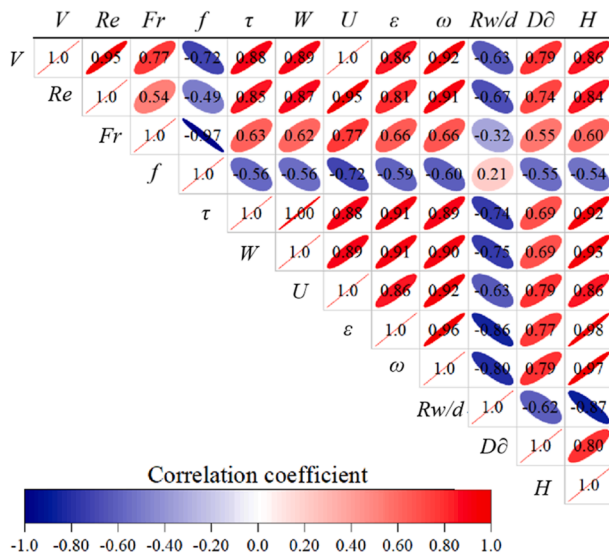


Fig. 12. Pearson correlation matrix between HHP and rill morphological indicators.

Table 5

Results of Grey relational analysis between hydraulic and hydrodynamic parameters and soil detachment rate.

Hydraulic and hydrodynamic parameter	Grey relational coefficient	Rank
V	0.7663	4
Re	0.8116	3
Fr	0.7186	6
f	0.6257	7
τ	0.8743	2
W	0.9056	1
U	0.7662	5

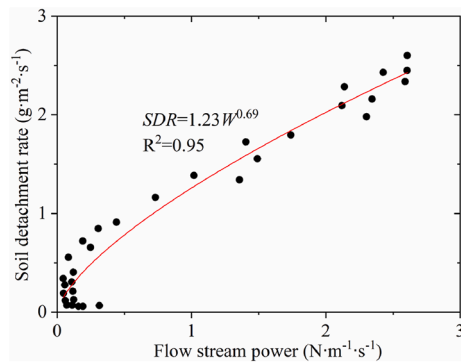


Fig. 13. Relation between soil detachment rate and flow stream power.

Overall, the variations in runoff and sediment yield under different rainfall intensities reflect the combined effects of rainfall energy, infiltration dynamics, and surface runoff generation. Higher rainfall intensities produce greater runoff volumes, shorter infiltration times, and more vigorous erosive forces, all of which accelerate rill formation and sediment transport. These findings align with established understanding of rainfall–runoff–erosion interactions (Nearing et al., 1997; Romkens et al., 2002) and provide essential insights into the hydrological and geomorphic processes driving rill initiation and development on hillslopes.

4.2. Rill development and morphological characteristics

The rill development processes under the four rainfall intensities (Fig. 14) demonstrate that rainfall intensity exerts a clear control on the rate of rill formation and the resulting morphological characteristics. Although all the experiments followed the general sequence of splash erosion, sheet flow, micro-depression formation, rill initiation, and the subsequent expansion of rill networks, the timing of rill appearance, the pace of development, and the degree of network complexity varied distinctly among rainfall intensities. At the beginning of the experiments, the nearly planar slope surface facilitated the formation of sheet flow. As microtopographic variations amplified local shear stress, small pits developed on the slope, marking the onset of rill erosion. Under higher rainfall intensities (90 and 120 mm/h), the greater flow energy and stream power led to earlier rill initiation, faster development, and the formation of more complex rill networks. In contrast, lower rainfall intensities (30 and 60 mm/h) produced weaker flow energy, resulting in later rill initiation and simpler network structures. The multi-temporal high-resolution terrain models obtained from the close-range photogrammetric system (Fig. 15) captured these dynamic processes in detail, reinforcing the conclusion that rainfall intensity is a key factor governing rill development.

The rill evolution processes results clearly demonstrate that rainfall intensity exerts a strong control on the pace and pattern of rill development. Under lower rainfall intensities (30 and 60 mm/h), rills initiated later and evolved into relatively simple and sparsely connected networks (Fig. 14). In contrast, higher intensities (90 and 120 mm/h) triggered earlier rill initiation and more rapid expansion, forming complex and highly connected networks (Fig. 14). These observations are consistent with the expected hydrodynamic response to increasing rainfall intensity and highlight the advantage of the DCRPS photogrammetric system, which enables continuous and high-resolution monitoring of rill evolution throughout the rainfall event.

The quantitative morphological indicators used in the current study, including rill density, splitting degree, fractal dimension, and geomorphic entropy, effectively characterize the complexity and spatial organization of rill networks. Their overall increase with rising rainfall intensity and rainfall duration (Figs. 8 and 9) reflects the transition of rill systems from simple channels to more intricate networks, consistent with the conclusion that enhanced flow energy promotes more rapid and complex rill evolution. Rill density describes the proliferation and spatial extent of rill channels, the splitting degree reflects the degree of branching within the network, while fractal dimension and geomorphic entropy capture the scale-dependent complexity and organizational state of the system. At lower rainfall intensities, the slow increase in these indicators corresponds to limited erosion energy and relatively weak rill development. At higher intensities, the rapid rise in these metrics indicates enhanced incision capacity and more complex morphological evolution. These observations are in line with previous findings on the nonlinear development of rill systems under rainfall-driven erosion processes (Ni et al., 2018; Zhang et al., 2017) and support the conclusions drawn in the current study.

The regression analysis results of Eq. (17) indicated a significant linear relation between the morphological indicators of geomorphic entropy, and fragmentation degree and sediment yield (Table 4). The change in rill morphology had an important impact on sediment yield. Based on the variation of the CQI with rainfall

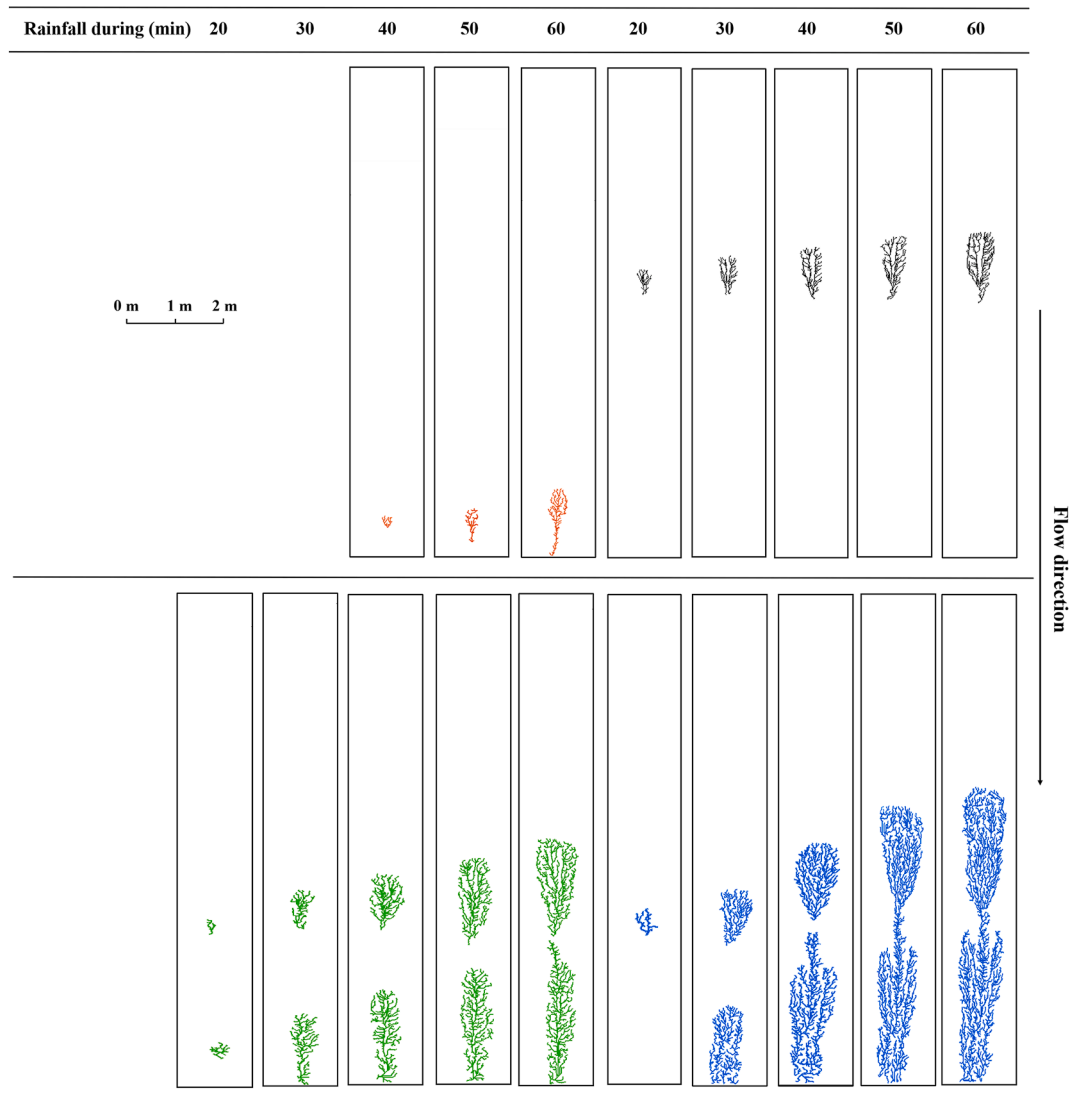


Fig. 14. Rill development networks with time. Orange, black, green, and blue figures represent rainfall intensities of 30, 60, 90, and 120 mm/h, respectively. Numbers (0–2 m) indicate the downslope distance from the plot top.

duration, it was observed that higher rainfall intensities corresponded to a faster increase in CQI values and more rapid changes in rill morphology. This nonlinear trend can be attributed to changes in runoff energy during rainfall events. As rainfall intensity increases, both runoff volume and stream power rise, resulting in greater scouring forces and increased sediment yield (Fig. 10). However, the energy consumed by runoff erosion, sediment detachment, and transport processes also increases with sediment yield, which eventually slows down the rate of morphological change. The interplay between increasing runoff energy and sediment yield leads to the observed nonlinear evolution of rill morphology with rainfall.

Previous studies (e.g., Ni et al., 2018; Zhang et al., 2019) also applied the CQI to quantify rill morphology. However, in those studies, the CQI was derived from static morphological parameters measured only after rainfall events under specific experimental conditions. In contrast, the CQI applied in the current study incorporates dynamic morphological data captured at multiple time intervals throughout the rainfall process, allowing for a more detailed and continuous representation of rill evolution. Therefore, the CQI developed in this study provides a more robust and

accurate framework for quantifying and characterizing rill morphology under varying rainfall intensities.

Overall, the results presented in Sections 4.1 and 4.2 indicate that rainfall intensity functions as the fundamental driver linking runoff generation, sediment yield, and rill development. Higher rainfall intensities produce greater runoff volumes and stronger flow energy, which enhance soil detachment and increase sediment yield. At the same time, the intensified hydrodynamic forcing accelerates rill initiation, deepens incision, and promotes the formation of more complex rill networks. Therefore, the increases in sediment yield and the progressive organization of rill morphology should be viewed as two manifestations of the same rainfall-driven erosion mechanism rather than independent phenomena. This integrated understanding helps unify the interpretations of runoff–sediment processes and rill evolution under varying rainfall intensities.

4.3. Hydraulic and hydrodynamic mechanisms of rill erosion

Water flow is the fundamental driving force behind the formation of rills (Nearing et al., 1988), and the processes of soil

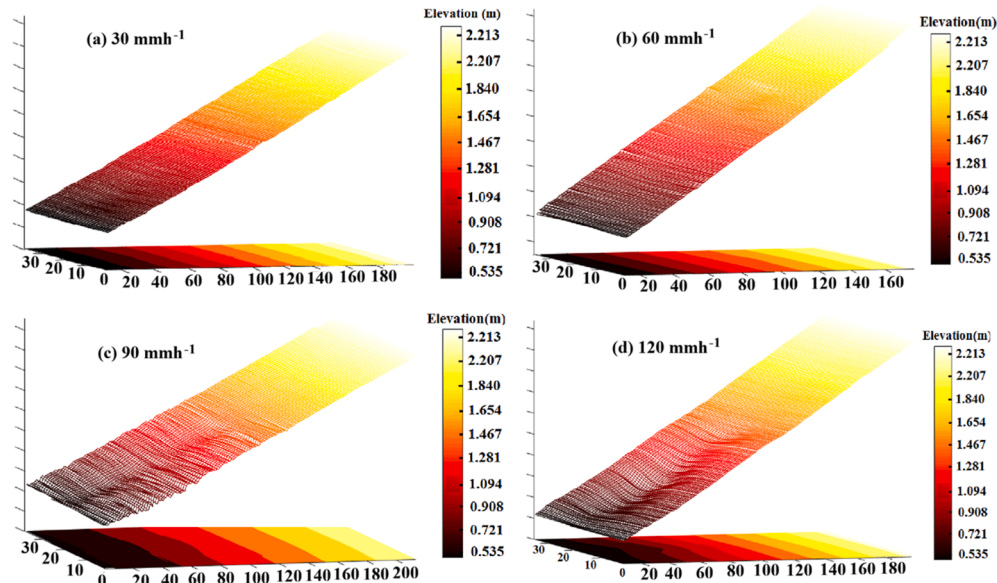


Fig. 15. Digital microtopography models at different rainfall intensities.

detachment and transport are essentially driven by energy consumption (Shen et al., 2016). The rill morphology and SDR are significantly correlated with HHP (Fig. 12). Therefore, investigating the dynamic mechanism of rill erosion is of great significance in revealing the mechanism of rill development. Furthermore, the differences in HHP between different rainfall intensities indicate that changes in rainfall intensity substantially affected by HHP (Fig. 11). The greater the rainfall intensity, the greater the energy possessed by the water flow, and the stronger the disturbance on the slope surface. After the appearance of rills on the slope, the water flow state transformed from thin-layer slope flow to concentrated rill flow, causing essential changes in hydrodynamic and erosion characteristics, providing sources of water flow and sediment for rill erosion by bringing soil particles into the rills (Li, Chen, Tao, et al., 2022). Moreover, the formation of rills provides pathways for sediment transportation. Compared to hillslope flow, rill flow can carry and transport more sediment, thus shaping complex landforms. Therefore, the erosion characteristics and development morphology of rills largely depend on the hydrodynamic characteristics of flow (Lou et al., 2022).

Notably, among all HHP, W is the best parameter to describe rill morphology (Fig. 12), and it exhibits a significant power function relation with the morphological parameters of H , D_0 , ε , and ω . In addition to rill morphology, W is also the best parameter for characterizing SDR. The denudation capacity is synchronized with the change of W , and it has also been noted that SDR increased exponentially with W (Fig. 13). This result is consistent with the conclusion obtained by Nearing et al. (1999), who proved that the stream power could more accurately simulate the soil separation process through field experiments. The denudation capacity was defined as a function of the unit stream power in the European EUROSEM and LISEM erosion models (Govers et al., 1990). In this study, W is more suitable for predicting soil erosion than τ and U , which agrees with the findings of Zhang et al. (2005). Although there is no significant difference between the conclusions of the current experiments and previous studies, the erodibility coefficient represented by the equation is significantly different. Such differences in coefficients are mainly attributed to variations in soil texture, structural stability, and slope characteristics across different experimental conditions, rather than differences in the

underlying hydrodynamic mechanism. In the current loess slope experiments, the distinct erodibility coefficient in the W -SDR relation reflects the unique properties of loess, including its aggregation behavior and high susceptibility to concentrated flow erosion. During the erosion process, the hydrodynamic characteristics, rill morphology, and sediment yield on the slope form a dynamic coupling system that is interrelated and mutually constrained (Sun et al., 2013). Therefore, elucidating the relation between rill hydrodynamics, morphological development, and sediment transport is essential for advancing the understanding of rill formation and evolution on slopes.

The superiority of W over τ and flow velocity in describing rill morphology evolution can be explained by its more complete representation of the erosive energy available for soil detachment. Unlike τ , which primarily depends on local hydraulic radius and slope and flow velocity, which reflects only the kinematic aspect of flow, W integrates both discharge and flow velocity, thereby quantifying the rate of energy expenditure along the flow path. This makes W more sensitive to the concentrated flow conditions that drive rill incision, branching, and network expansion. The current experimental results also support this interpretation: W shows the strongest functional relation with rill depth, fragmentation degree, geomorphic entropy, and other morphological indicators, whereas τ and U exhibit weaker or less consistent associations. In addition, the exponential increase of SDR with W further demonstrates that stream power more accurately reflects the hydrodynamic forcing that governs rill evolution. These findings are consistent with previous studies (Govers et al., 1990; Nearing et al., 1999; Zhang et al., 2005), confirming that W is the most reliable hydrodynamic indicator for characterizing rill erosion on slope surfaces.

4.4. Suggestions for future studies

In the current study, only one variable factor of rainfall intensity was considered to investigate the evolution process of a soil erosion landscape. Since the purpose of the current study was to explore the erosion characteristics in the process of dynamic evolution, the setting of multiple variable factors was not crucial to achieve the study goal. The current study provides a reference for

future research on the erosion process, so more experiments will be done for multiple controllable factors in the future, such as raindrop size distribution; kinetic energy of simulated rainfall; and different soil types, slope lengths, and slope gradients. In addition, due to the difficulty of image acquisition technology, the digital close-range photographic observation technology utilized in the current study only allows the erosion tests to be done on a bare slope, and it cannot monitor the dynamic evolution characteristics of the erosion morphology with vegetation coverage. In the future, the observation system will be improved to enable it to penetrate vegetation on the soil surface and remove the surface vegetation to obtain the information of erosion deformation. If successful, the experiments will be applied for the field conditions with various soil types and landscapes. The technology also is potentially applicable elsewhere in other regions.

5. Conclusions

The current study did four rainfall simulation experiments to investigate how rainfall intensity controls runoff and sediment yield, rill morphological evolution, and associated hydrodynamic mechanisms. Using a digital close-range photogrammetry system, continuous and high-resolution monitoring of rill development during rainfall was achieved, enabling quantitative analysis of erosion dynamics. Runoff rate and sediment yield both increased markedly with rainfall intensity, and higher intensities (90 and 120 mm/h) produced substantially stronger surface disturbance than lower intensities. Morphological indicators including rill density, splitting degree, fractal dimension, and geomorphic comentropy, showed consistent increases with rainfall duration and intensity, whereas the width–depth ratio decreased. Among these indicators, geomorphic comentropy and splitting degree exhibited the strongest association with sediment yield and were, therefore, used to construct a comprehensive quantitative indicator (CQI). The CQI increased exponentially during rainfall, with faster growth under higher intensities, capturing the progressive organization and complexity of rill networks. Among all hydrodynamic variables, stream power (W) proved to be the most effective parameter for characterizing both erosion intensity and morphological adjustment. Overall, the results clarify the quantitative relation linking rill morphology, hydrodynamic conditions, and sediment detachment, providing new insight for process-based understanding of rill erosion and for improving erosion prediction on hillslopes.

CRediT authorship contribution statement

Yanmin Jiang: Writing – original draft, Visualization, Methodology, Investigation, Conceptualization. **Haijing Shi:** Supervision, Project administration, Funding acquisition. **Xihua Yang:** Formal analysis, Data curation, Conceptualization. **Minghang Guo:** Resources, Funding acquisition. **Weihua Fang:** Writing – review & editing, Methodology. **Junfeng Shui:** Software, Investigation. **Youfu Wu:** Writing – review & editing, Investigation. **David Paull:** Writing – review & editing. **Ye Wang:** Methodology, Investigation.

Declaration of competing interest

We declared that we have no conflicts of interest to this work. This manuscript has not been published or presented elsewhere in part or in entirety and is not under consideration by another journal. All study participants provided informed consent, and the study design was approved by the appropriate ethics review board. We have read and understood your journal's policies, and we

believe that neither the manuscript nor the study violates any of these.

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